

“I look like a gorilla, but don’t move like one!”: Impact of Avatar-Locomotion Congruence in Virtual Reality

Omar Khan*
Department of Computer Science
University of Calgary

Hyeongil Nam†
Department of Electrical and Software
Engineering
University of Calgary

Kangsoo Kim‡
Department of Electrical and Software
Engineering
University of Calgary

ABSTRACT

As virtual reality (VR) continues to expand, particularly in social VR platforms and immersive gaming, understanding the factors that shape user experience is becoming increasingly important. Avatars and locomotion methods play central roles in influencing how users identify with their virtual representations and navigate virtual spaces. However, little is known about the impact of congruence between these two factors. We conducted a user study with 30 participants, employing two avatar types (human and gorilla) and two locomotion methods (human-like arm-swing and gorilla-like arm-roll), to assess the effects of avatar-locomotion congruence. Our results indicate that congruent avatar-locomotion conditions enhance avatar identification and user experience.

Index Terms: Virtual reality, avatars, appearance, locomotion, avatar identification, usability, user experience, gamification.

1 INTRODUCTION

The increasing use of virtual avatars underscores the importance of understanding avatar identification, particularly as it becomes integral to both social and immersive experiences in virtual reality (VR) [3]. In VR gaming, users are often embodied as virtual characters ranging from human-like figures to animals or other creatures. Key factors, such as avatar appearance and locomotion methods, play significant roles in determining the success and enjoyment of VR experiences.

While some studies have highlighted the importance of the interrelation between locomotion methods and avatars [6], we did not find previous research clearly addressing the impacts of avatar-locomotion congruence. For example, Gao et al. [4] found that avatar appearance has minimal impact on locomotion experience, and Dewez et al. [2] found no significant difference in embodiment across locomotion techniques.

In this extended abstract, we explore an overlooked aspect of this inter-relation, by assessing how *congruence/incongruence* between the two factors impacts user experience and avatar identification in VR. For instance, users might be embodied as a human avatar, even though their locomotion method may not intuitively match human-like movements. This mismatch between avatar appearance and movement could profoundly affect how users identify with their avatars and their overall VR experience.

To address this research gap, we designed and conducted a user study. In the study, we prepared two avatar appearances (i.e., gorilla and human avatars) and two locomotion methods (i.e., arm-roll and arm-swing) to investigate the effects of these factors (see Figure 1). Our motivation for the arm-roll locomotion is inspired by the VR

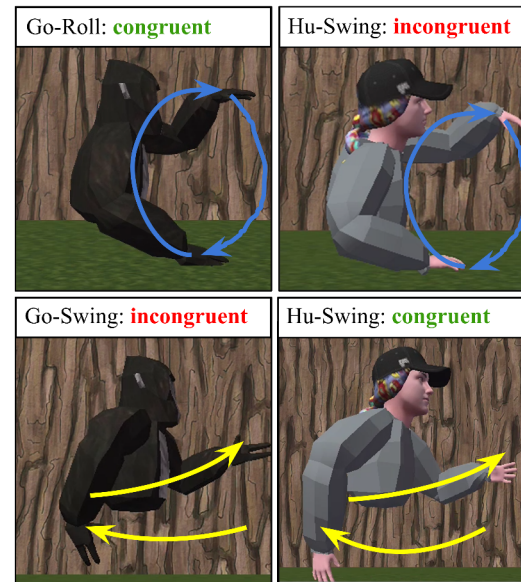


Figure 1: The four avatar-locomotion combinations in our study.

game, Gorilla Tag¹. The arm-roll motion resembles the movement of gorillas and other quadrupeds, which is congruent with gorilla avatars. Conversely, the arm-swing locomotion resembles the way humans walk in the real world, making them congruent with human avatars.

2 MATERIALS: VIRTUAL OBSTACLE GAME

We created a program for the Meta Quest 3 head-mounted display using the Unity game engine (v2021.3.29f1), consisting of a virtual obstacle game. The environment consists of a starting area and an ending area, with a series of obstacle courses connecting the two. We added nature props², to decorate the environments, and stacked crates³ to serve as obstacles. There are seven mirrors placed in the virtual environment so that the users could see their avatars during the experience. To create the gorilla avatar, we used a gorilla avatar model⁴ from the web, similar to the one used in Gorilla Tag. The human avatar model was made using the avatar creation platform, Ready Player Me, and modified in Blender (v4.1.1) to match the body shape of the gorilla model. Users could choose one of the five skin tones for their avatar, allowing them to select a tone similar to

¹Gorilla Tag: <https://www.meta.com/experiences/gorilla-tag/4979055762136823/>

²Nature props: <https://assetstore.unity.com/packages/3d/environments/landscapes/low-poly-simple-nature-pack-162153> (Accessed 2024-09-12)

³Crates: <https://assetstore.unity.com/packages/3d/props/low-poly-crates-80037>

⁴Gorilla avatar model: <https://sketchfab.com/3d-models/gorilla-tag-gorilla-textured-rigged-colour-43404d61ecf74111993770ee437f913e> (Accessed 2024-09-12)

*e-mail: omar.khan2@ucalgary.ca

†e-mail: hyeongil.nam@ucalgary.ca

‡e-mail: kangsoo.kim@ucalgary.ca (correspondence)

their own skin. To implement the arm-roll locomotion method, we adopted the open-source Gorilla Tag locomotion code.⁵ By pulling, pushing, or dragging their hands against the virtual ground, the user exerts a simulated force against the virtual ground, thereby generating momentum. For arm-swing locomotion, we made a custom script which allows users to move around by swinging their arms in a natural walking motion. To move around the virtual environment, the user first selects a direction on the joystick on their right controller, then swing their arms in a natural walking motion to move in the selected direction.

3 STUDY DESIGN

We recruited 30 participants (13 female, 17 male, age $M = 21.7$, $SD = 3.2$, range: 19–35) for a within-subjects study approved by the University of Calgary Research Ethics Board (REB24-0729). Our study design had two independent variables—avatar type and locomotion method—resulting in four experimental conditions: *Go-Roll* (gorilla avatar with arm-roll locomotion, Figure 1a), *Hu-roll* (human avatar with arm-roll locomotion, Figure 1b), *Go-Swing* (gorilla avatar with arm-swing locomotion, Figure 1c), and *Hu-Swing* (human avatar with arm-swing locomotion, Figure 1d). Before the study session, they filled out a pre-study questionnaire to provide some demographic information. They learned the two locomotion methods in a training session without any avatar embodiment. After the learning phase, every participant experienced all four conditions in a counterbalanced order, spending four minutes per condition navigating the obstacle course.

After each session, we measured the participants' avatar identification and user experience. For avatar identification, we used a questionnaire developed by Van Looy et al. [7], which consists of three sub-dimensions: wishful identification, embodied presence, and similarity identification. Our measure of user experience consists of four sub-dimensions: enjoyment, satisfaction, ease of learning, and task load. We adopted the enjoyment scale from the Intrinsic Motivation Inventory [1], the satisfaction and ease of learning scales from the USE questionnaire [8], and the NASA TLX [5] for task load. Participants received an \$15 CAD of Amazon gift card as compensation for participating in the study. We hypothesized that participants would experience stronger avatar identification, and more positive user experience, in congruent avatar-locomotion conditions.

4 RESULTS

The questionnaire data was analyzed using IBM SPSS (v29.0.1.1). We conducted two-way repeated measures ANOVA tests at a 95% confidence level. All necessary assumptions for the tests were verified and met, and Bonferroni corrections were applied to the post hoc pairwise comparisons.

Avatar Identification Avatar identification was measured across three dimensions: wishful identification, embodied presence, and similarity identification. Significant interaction effects were observed between avatar appearance and locomotion method for wishful identification ($F(1,29) = 11.792$, $p = 0.002$, $\eta_p^2 = 0.289$), embodied presence ($F(1,29) = 14.612$, $p < 0.001$, $\eta_p^2 = 0.335$), and similarity identification ($F(1,29) = 10.144$, $p = 0.003$, $\eta_p^2 = 0.259$). Pairwise comparisons revealed higher scores for congruent conditions: *Go-Roll* ($M = 3.827$, $SE = 0.214$) and *Hu-Swing* ($M = 3.840$, $SE = 0.282$) in wishful identification; *Go-Roll* ($M = 5.225$, $SE = 0.188$) and *Hu-Swing* ($M = 4.992$, $SE = 0.274$) in embodied presence; and *Hu-Swing* ($M = 4.567$, $SE = 0.276$) in similarity identification. These results suggest that congruent pairings, such as the gorilla avatar with arm-rolling (*Go-Roll*) and

the human avatar with arm-swinging (*Hu-Swing*), lead to stronger avatar identification across all measures, which supports our hypothesis.

User Experience User experience was evaluated through enjoyment, satisfaction, ease of learning, and task load. Enjoyment showed a significant main effect of avatar appearance ($F(1,29) = 7.502$, $p = 0.010$, $\eta_p^2 = 0.206$) and an interaction effect ($F(1,29) = 7.552$, $p = 0.010$, $\eta_p^2 = 0.207$). Participants reported the highest enjoyment in the *Go-Roll* condition ($M = 5.914$, $SE = 0.108$). Satisfaction showed a significant main effect of locomotion method ($F(1,29) = 10.425$, $p = 0.003$, $\eta_p^2 = 0.264$) and a strong interaction effect ($F(1,29) = 30.064$, $p < 0.001$, $\eta_p^2 = 0.509$), with *Go-Roll* rated highest ($M = 5.928$, $SE = 0.142$). Usability measures revealed significant effects for ease of learning ($F(1,29) = 9.171$, $p = 0.005$, $\eta_p^2 = 0.240$) and task load ($F(1,29) = 5.850$, $p = 0.022$, $\eta_p^2 = 0.168$). Participants found arm-rolling (*Go-Roll*: $M = 5.928$, $SE = 0.142$) easier to learn, while task load was lowest in the *Hu-Swing* condition ($M = 24.583$, $SE = 2.780$). These findings indicate that congruent avatar-locomotion pairings improve enjoyment, satisfaction, ease of learning, and reduce perceived task load, which supports our hypothesis.

5 CONCLUSION

This extended abstract explored how the congruence/incongruence between avatar appearance and locomotion influences avatar identification and user experience in VR. Through a within-subjects experiment with two avatar appearances (i.e., human and gorilla) and two locomotion methods (i.e., arm-swing and arm-roll), we found that congruence significantly enhances avatar identification, enjoyment, satisfaction, and ease of learning. These findings highlight the importance of aligning avatar appearance with locomotion methods to enhance user immersion and overall VR experience, with implications for the design of immersive gaming, social VR experiences, and remote physiotherapy applications.

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⁵Gorilla Tag locomotion source code: <https://github.com/Another-Axiom/GorillaLocomotion> (Accessed 2024-09-12)